Lab 13 - Layer Height and Top Thickness - Surface Quality

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 03: Process Improvement for AM TSC objective: A5 - optimise layer height and top thickness for surface quality vs speed

Goal

Slice the same model at 0.1 mm, 0.2 mm and 0.3 mm layer heights, adjust top thickness to a constant, and compare surface quality against print time.

What you'll build

Sliced (and optionally printed) evidence of the layer-height ladder: finish vs time.

Tools: Ultimaker Cura, cube10.stl, sloped-cube.stl, FDM printer

Steps

1. Open cube10.stl and slice at 0.2 mm layer height; record the time.

```
Quality > Layer Height - 0.2 mm
```

2. Slice at 0.1 mm; note the doubled layer count and time.

```
Layer Height - 0.1 mm
```

3. Slice at 0.3 mm; note the coarser stairs and the reduced time.

```
Layer Height - 0.3 mm
```

4. Keep top thickness constant at 0.8 mm - so the TOP LAYER COUNT changes with layer height (8 / 4 / 3 layers).

```
Top/Bottom > Top Thickness - 0.8 mm
```

5. In Preview, compare side-wall stair-stepping and the roof at each height.

```
Preview > compare walls + roof
```

6. Print the 0.1 mm and 0.3 mm cubes and compare finish by eye and touch.

```
print both
```

7. Repeat the three slices with sloped-cube.stl and zoom into the sloped face; this geometry makes the staircase effect visible in Preview before printing.

```
Open sloped-cube.stl > Slice at 0.1 / 0.2 / 0.3 mm > compare
```

Evidence to capture

- Three-row layer-height table with layer count, print time and observed stair-stepping.

- Matched Preview screenshots of the sloped face at 0.1, 0.2 and 0.3 mm.

Test it

You can rank the layer heights by finish and time, point to visible stair-stepping on sloped-cube.stl, and explain why constant top thickness-not a fixed top-layer count-keeps the comparison fair.

Files in this folder

- cube10.stl - Core sample model used by the lab procedure.
- sloped-cube.stl - Sloped-face benchmark that makes layer-height stair-stepping easy to compare.